

Supply Chain Optimization and Simulation for Blast Freezer Operations

Client: PFG Group, Inc.

Executive Summary

PFG Group, Inc. is a third party logistics provider based in Madison, Wisconsin specializing in blast freezing poultry and dairy products. Blast freezing is a capital intensive operation, making effective utilization of freezer capacity critical to operational performance. Currently, PFG operates with approximately 35–40% of pallets in process, well below its target range of 70–80%. The company seeks a decision support framework to guide pallet placement decisions upon truck arrival and to assess operational performance over a monthly horizon.

This project was conducted by graduate students in the INFO 645 Prescriptive Analytics course at Virginia Commonwealth University during the Fall 2025 semester. The work was completed in two phases over the course of the term. Phase 1 focused on developing an optimization model to determine how incoming pallets should be assigned to freezer cells. Phase 2 extended this work by developing a simulation model to represent operational uncertainty and evaluate system performance over a 30 day period.

The integrated simulation optimization framework achieved an average freezer utilization of approximately 92.6% over the 30 day simulation, substantially higher than current operational levels. The system processed approximately 7,000 pallets during the

simulated month and frequently operated at or near full capacity. Based on these findings, we recommend transitioning from fixed zone based pallet assignments to a dynamic, cell level assignment policy supported by optimization and simulation based capacity planning.

1. Objective in Words

The objective of this project is to develop a decision support framework that determines how incoming pallets (SKUs) should be assigned to blast freezer cells to improve utilization of freezer capacity and overall throughput. The optimization model seeks to maximize a priority based matching score that aligns pallets requiring longer freezing times with freezer cells that have lower remaining occupied time, subject to freezer capacity and assignment constraints. The objective function does not represent profit or cost, but rather a relative measure of alignment between pallet freezing requirements and available freezer capacity.

2. Data and Assumptions

The Phase 1 optimization model uses deterministic snapshot data including SKU specific freeze times, remaining occupied time for each freezer cell and a total freezer capacity of 600 cells. Where detailed freeze time data were unavailable, benchmark values and averages derived from historical sensor data were used.

Phase 2 incorporates stochastic inputs to capture operational uncertainty, including

daily truck arrivals, pallets per truck and SKU mix. Probability distributions were fitted to historical data and used to generate daily demand over a 30 day simulation horizon.

3. Decision Variables

Let $x(c,s) = 1$ if pallet s is assigned to freezer cell c and 0 otherwise, for all pallets s and freezer cells c .

4. Optimization Model Formulation

The optimization model maximizes a priority based matching score defined as the ratio of pallet freeze time to remaining freezer cell occupied time. The model is subject to constraints ensuring that each freezer cell holds at most one pallet, each pallet is assigned to at most one freezer cell and all decision variables are binary.

5. Simulation Model Description

The Phase 2 simulation model represents daily blast freezer operations over a 30 day period. Each simulated day updates freezer states, generates stochastic arrivals and executes the Phase 1 optimization model using the current system state. This integration ensures that simulation outputs directly serve as inputs to the optimization model.

6. Results and Findings

Phase 1 results demonstrate optimal pallet to cell assignments based on the defined priority based objective function. The resulting objective value represents a relative matching score rather than a monetary quantity.

When embedded within the 30 day simulation, the model achieved an average freezer utilization of approximately 92.6%. The system processed approximately 7,000 pallets and operated at or near full capacity on several days, indicating that demand rather than freezer capacity is the primary operational constraint under higher utilization levels.

Conclusions and Recommendations

The results indicate that PFG can significantly improve freezer utilization and throughput without additional capital investment by adopting a dynamic, cell level pallet assignment policy supported by optimization. We recommend using the simulation framework as an ongoing capacity planning tool to support customer scheduling, acceptance decisions and operational planning.

Colab Link:

<https://colab.research.google.com/drive/1uKJcfb5mCLBmODr7XRUq5yV31GnTi>

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